

Ground-to-Cable Strain Transfer in Unburied DAS on Earth and the Moon

S. Probst¹, M. Serra Garcia², J.O.A. Robertsson¹, C.M. Donahue³

¹Department of Earth and Planetary Sciences, ETH Zurich, Zurich, Switzerland

²AMOLF, Amsterdam, The Netherlands

³Earth and Environmental Sciences, Los Alamos National Laboratory, NM, USA

Abstract

Distributed Acoustic Sensing (DAS) measures dynamic strain along a fiber-optic cable, offering a robust, densely-sampled alternative to traditional seismic sensors. To ensure good ground-to-cable coupling, cables are typically buried in a shallow trench. Unburied surface deployments are attractive for rapid-response terrestrial applications as well as extraterrestrial missions, such as on the Moon, where burial is impractical. However, unburied DAS often suffers from severely degraded signal quality, due to poor strain transfer from ground to cable. The physical mechanism responsible remains unknown. Here, we identify bending stress relief as a mechanism sufficient to explain this loss: suspended cable segments accommodate ground strain by bending rather than by stretching or compressing, reducing the measurable axial strain that reaches the fiber. We develop the first analytical and numerical model of unburied DAS coupling, representing the draped cable as a series of suspended segments between discrete ground contact points, to explain and quantify the bending stress relief mechanism. We introduce a dimensionless parameter Θ , set by the ratio of the cable's initial gravity-induced sag to its radius. Once a segment's sag exceeds a quarter of the cable's radius, ground displacement is absorbed by bending rather than being transferred as measurable axial strain. This framework predicts how mechanical properties, cable dimensions, pretension, and gravity affect coupling efficiency and provides quantitative guidelines for optimizing cable design and deployment strategies on both Earth and the Moon.

1 Introduction

DAS is a fiber-optic sensing technique that measures the dynamic axial strain along a cable by analyzing phase shifts in Rayleigh-backscattered laser light (e.g., Hartog, 2017; Lindsey & Martin, 2021). DAS records the elongation or shortening of an optical fiber, leading to an inherently one-dimensional measurement of strain inline with the sensing cable.

Over the past decade, DAS has been adopted widely for borehole seismic applications such as vertical seismic profiling (Daley et al., 2013; Mateeva et al., 2014), and increasingly for surface deployments in which the cable is placed in a shallow trench and coupled to the ground by burial and compaction (Abukrat et al., 2023; Hubbard et al., 2022).

Burial serves two purposes: it shields the cable from wind-induced noise and ensures continuous mechanical contact with the ground, so that the strain is transferred from the surrounding medium onto the cable.

Empirical results widely show that unburied DAS deployments, lacking this continuous contact, exhibit degraded signal quality relative to buried configurations. An et al. (2023) demonstrated this by comparing DAS recordings of traffic-induced vibrations on a road using the same cable in two configurations, cemented to the surface and surface-draped, finding that the latter recorded signals with significantly weaker amplitudes and lower signal-to-noise ratio (SNR). Similarly, Harmon et al. (2022) compared surface coupling strategies in a grassy field and found that a surface-draped cable could only capture hammer-source energy within approximately 10 m, while weighted configurations more than doubled this range. Zandanel et al. (2026) compared buried and unburied deployments in a lunar regolith simulant and found that unburied cables were able to record earthquakes but with lower amplitude compared to the buried configuration.

Part of this amplitude reduction can be attributed to atmospheric noise: wind exerts mechanical forces on the cable which contaminates the seismic signal (Hudson et al., 2021; Probst et al., 2026; Viens et al., 2025). However, poor signal quality persists even in wind-free environments: Probst et al. (2026) showed in controlled laboratory experiments that unburied cables recorded substantially lower amplitudes than buried cables even without any fan-induced wind. These observations indicate that poor ground coupling, and not atmospheric noise alone, is a fundamental challenge of unburied DAS.

Despite this challenge, unburied deployments are attractive, and in some settings indispensable. Burial is costly, time-consuming and requires excavation infrastructure that is unavailable or impractical in many environments. This makes unburied deployments a necessity, for example, in rapid-deployment scenarios such as recording aftershocks following a large seismic event.

Burial is also unfeasible in the challenging lunar environment, where the fine-grain and abrasive regolith poses large risks to infrastructure. DAS has nonetheless been proposed for deployment on the Moon because its robustness and dense spatial sampling make it an attractive alternative to single-station seismometers. On the Moon, a rover could unroll kilometers of fiber-optic cable directly onto the surface, and record moonquakes, meteorite impacts, and active sources (Wu et al., 2024; Zhai et al., 2024). Understanding whether and how such a deployment can transfer ground strain to the fiber is therefore important to establish.

A cable resting on or buried in the ground does not transmit ground strain to the fiber core perfectly. One component of this transfer, which we term cable-to-fiber transfer, concerns how strain is transmitted through the multiple concentric layers of a cable’s cross-section (jacket, buffer, coating, cladding, and fiber core). This process is governed by shear-lag mechanics. Softer intermediate layers, such as the gel inside loose-tube telecommunications cables, attenuate the strain before it reaches the fiber core (Falcetelli et al., 2020; Hubbard et al., 2022; Reinsch et al., 2017; Tan et al., 2021). Empirical comparisons confirm this by consistently showing that tight-buffered cables yield higher strain transfer rates than loose-tube cables (Castongia et al., 2017; Forbriger et al., 2024). A second, less well-understood step concerns how the seismic strain field in the ground is

transferred to the cable’s outer surface in the first place — what we call ground-to-cable transfer. For buried deployments, this step has been studied analytically and numerically. Zhang et al. (2020) developed a theoretical framework for the ground-backfill-cable interaction for distributed strain sensing in boreholes. Celli et al. (2023) studied cable-ground coupling for DAS using a spring model, finding that the stiffness of the material immediately surrounding the cable is the dominant factor, leading to amplitude amplification and phase delays.

Together, these two steps govern how the ground strain is first transmitted to the cable outer surface, and then propagated inward through the cable layers to the fiber. For buried DAS, the physical mechanisms of both steps are reasonably well characterized. This study concerns unburied cables, for which the first step in this chain, the ground-to-cable transfer, remains poorly understood.

With no overburden, the only force coupling the cable to the surface is its own weight. Contact is intermittent as the cable bridges over surface irregularities and touches the ground only at discrete points. The tangential force available to transmit axial seismic strain from the ground to the cable is limited by Coulomb friction at these contact points. Existing research on unburied DAS deployments is almost entirely empirical. Field experiments have shown that adding weights such as sandbags improves coupling (Harmon et al., 2022), though not consistently (Forbriger et al., 2024). Pressing a cable into snow improved coupling compared to draped deployment (Mjehovich et al., 2023).

Probst et al. (2026) found that thicker and stiffer cables perform better in unburied configurations and proposed bending stress relief as a key mechanism: a cable spanning contact points tends to bend rather than deform axially, so that the ground-induced axial strain is partially absorbed by cable bending rather than transmitted to the fiber. Thicker, stiffer cables resist this bending more effectively and could therefore record ground strain more reliably. However, this mechanism remained a qualitative hypothesis: it had not been tested or linked analytically to cable parameters.

A second potential mechanism reducing the strain transfer in unburied deployments is frictional slip between the cable and the ground at the contact points. As noted by Celli et al. (2023), for accelerations well below 1 g, an unburied DAS cable depends entirely on frictional contact for its coupling. If the amplitude of the ground-induced axial force exceeds the maximum static friction available at the contact points, the cable will slip and reduce the axial strain transmitted to the fiber. Both mechanisms, bending stress relief and frictional slip, are expected to depend on cable parameters including the bending stiffness EI (product of Young’s modulus E and area moment of inertia I), axial stiffness EA (product of Young’s modulus E and cross-section area A), mass per unit length $m = \rho A$, outer radius r , and the friction coefficient μ between the cable and the surface. Furthermore, both mechanisms are dependent on gravity, which is much lower on the Moon, at about one sixth of Earth’s gravity. It is therefore crucial to understand how gravity impacts these mechanisms, to predict the coupling in a lunar environment. At present, neither mechanism has been established as the physical explanation for the observed coupling losses, and no model exists that predicts how well a given cable couples to the ground in a surface deployment.

The goal of this paper is to address this gap by identifying and quantifying the physical mechanism governing ground-to-cable strain transfer in unburied DAS deployments, through an analytical and numerical model. We derive solutions for the axial strain transferred from the ground to the cable, modeling surface-draped cable segments as beams suspended between two contact points. We introduce a dimensionless number, a

ratio of the involved parameters, which predicts the strain transfer efficiency for a given deployment. Our results provide the first quantitative framework for understanding and optimizing unburied DAS coupling, with implications for cable design on Earth and on the Moon.

2 Analytical Strain Transfer Model

2.1 Treatment of Cable as Segments Suspended Between Ground Contact Points

An unburied, surface-draped cable is modeled as a sequence of suspended segments between contact points (Figure 1a). The spacing between ground-cable contact points depends on surface roughness and cable properties. This characteristic segment length would have to be measured for a given deployment and is here treated as an input to the model.

When strained, hanging segments might bend rather than deform axially. How the strain is transferred from the ground via the contact points to the cable itself is calculated using beam theory. The hanging cable segments are modeled as geometrically nonlinear Euler-Bernoulli beams which are deformed by the movement of their endpoints, the contact points with the ground. The following derivations and the numerical implementation quantify a single suspended segment. The recorded DAS response is the average over the gauge length, which spans many such segments of varying length. We comment on this gauge-length averaging in the discussion, and focus here on characterizing the behavior of an individual segment, which is the building block of this averaged response.

We consider the cable segment as a horizontal beam with mechanical properties: undeformed length L_0 , radius r , cross-sectional area $A = \pi r^2$, moment of inertia $I = \pi r^4/4$, Young's modulus E , and density ρ .

To allow for an analytic treatment of the problem the following assumptions are made. A cable segment with no inherent bend is assumed, which is fully straight in the absence of external forces. The contact points of the segment are assumed to be level (i.e., on the same height). The treatment is limited to an inline strain field with wavelengths much larger than the cable segment, such that the strain is essentially constant over the length of the segment. No slip at the contact points is assumed: the motion of the cable at the contact points is identical to the ground motion at these points. This also implies no interaction between adjacent suspended segments, since the behavior of each segment is fully dictated by the movement of its endpoints.

The hanging cable segment at rest is thus modeled as a beam with doubly-clamped endpoints subject to gravity (Figure 1b). When the endpoints are displaced, the cable deforms to accommodate the applied stress. The deformation is limited to two modes for this simplified treatment. The validity of this is seen in the later comparison to the numerical, exact solution. The two deformation modes are axial elongation q_l and the first flexural bending mode q_f (Figure 1c and d). The axial mode q_l is the change in the beam's length, while q_f is the maximum transverse deflection from the straight beam, which occurs at the midpoint. The transverse deflection is approximated as $w(x) = q_f \phi(x)$, where the first symmetric mode for a doubly-clamped beam is chosen as

$$\phi(x) = \frac{1 - \cos\left(\frac{2\pi x}{L}\right)}{2}, \quad (1)$$

with L the horizontal span between the endpoints. This satisfies the boundary conditions $w(0) = w(L) = 0$ (beam endpoints fixed), and $w'(0) = w'(L) = 0$ (beam endpoints orientation fixed), and gives $w(L/2) = q_f$ at the midpoint.

DAS primarily measures changes in optical path length along the fiber. In the present treatment, we assume that the DAS response is dominated by axial elongation of the cable segment and we neglect potential effects of fiber curvature on the optical measurement. Under this assumption, only the axial deformation mode contributes to the measured DAS signal while strain accommodated by bending is not recorded. The strain transfer efficiency describes the fraction of the axial strain which is transferred from ground to cable and is defined as $\frac{\delta q_l}{\delta L}$, where δq_l is the axial length change of the cable segment and δL is the change of the horizontal span.

2.2 Analytical Strain Transfer Efficiency

The system we consider starts from an equilibrium configuration, in which the suspended cable segment has sagged under its own weight until the gravitational and bending-resisting forces balance. When a seismic wave arrives, it displaces the segment's endpoints slightly, perturbing the equilibrium. The segment responds by partitioning the displacement between its two deformation modes, bending and axial elongation. Only the axial component is transferred to the fiber as measurable strain, so this split between modes is what determines the strain transfer efficiency. To determine the transfer efficiency, we first calculate the rest configuration under gravity alone, by equating the gravitational and bending-resisting forces. We then linearize the system about this sagged state and minimize the energy to calculate how a small endpoint displacement is partitioned between the two modes.

The arc-length of the beam is given by $S = \int_0^L \sqrt{1 + (w'(x))^2} dx$. Assuming small slopes ($|w'(x)| \ll 1$), expand to second order:

$$S \approx \int_0^L \left(1 + \frac{1}{2}(w'(x))^2 \right) dx = L + \frac{1}{2} \int_0^L (w'(x))^2 dx. \quad (2)$$

Substitute ($w(x) = q_f \phi(x)$):

$$S = L + \frac{1}{2} C_1 q_f^2, \quad C_1 = \int_0^L (\phi'(x))^2 dx = \frac{\pi^2}{2L}. \quad (3)$$

The arc-length of the beam must equal the undeformed beam length plus axial elongation $S = L_0 + q_l$. Thus, bending reduces the effective horizontal span for a given axial length:

$$L = L_0 + q_l - \frac{1}{2} C_1 q_f^2. \quad (4)$$

This geometric constraint is the only place where we distinguish between the undeformed material arc-length L_0 and the horizontal span L ; elsewhere, for the evaluation of the coefficients C_1 and C_2 and the energies below, we adopt the small-strain approximation $L_0 \approx L$.

Assuming small strains, the cable responds linear-elastically, and the energy of the axial

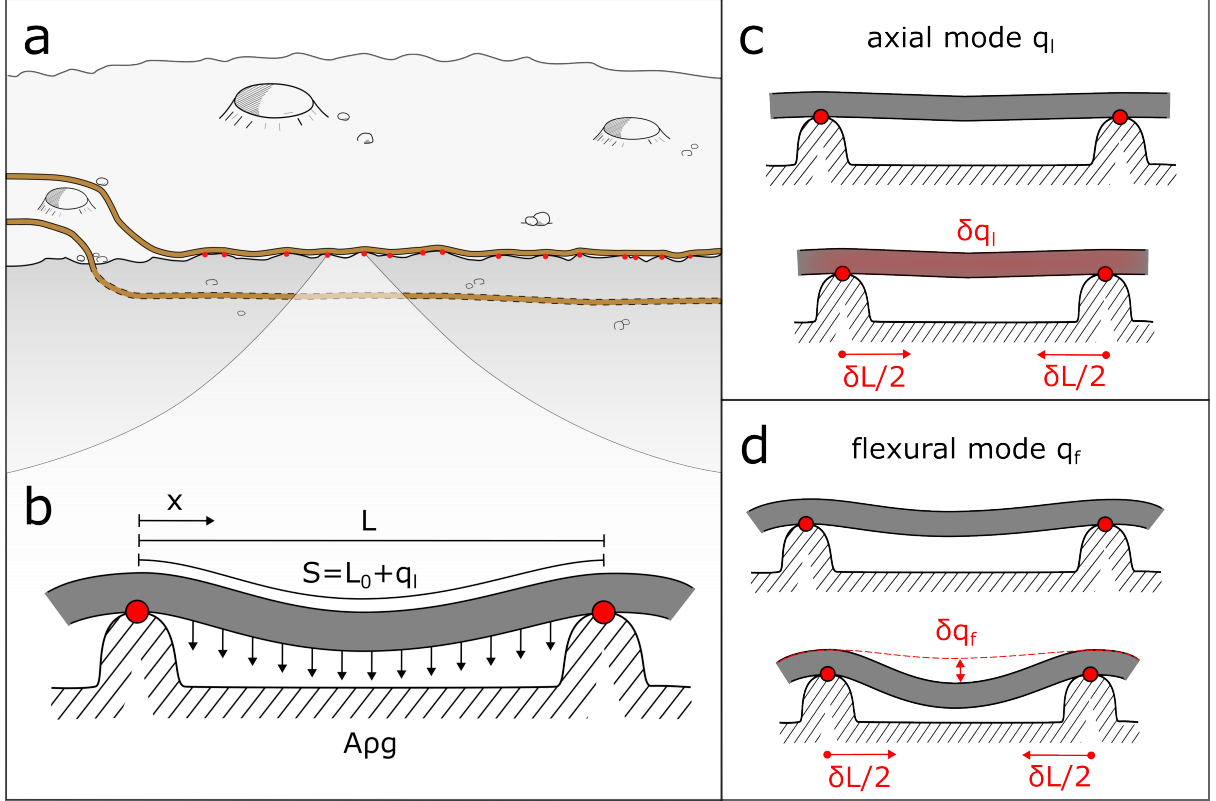


Figure 1: Unburied cables lack continuous ground contact, and form suspended segments between discrete contact points that accommodate applied ground strain through a combination of axial and flexural deformation modes. (a) Illustrative setup of a fiber-optic cable surface deployment in a lunar environment. (b) Zoom-in on a single suspended segment showing the variables used in the analytical derivations: the horizontal span between endpoints L , the arc-length S , the undeformed segment length L_0 , and the axial deformation q_l (change in arc-length). Gravity acts on the segment as a uniform distributed load ρAg , leading to sagging. When the segment endpoints are deformed by an applied ground displacement δL , this strain is partitioned into two principal deformation modes. (c) The axial mode, where the cable accommodates the displacement by physically changing its arc length. The axial strain from the ground is transferred to the cable and can be measured by DAS. (d) The flexural mode, where the cable bends to accommodate the displacement without changing its actual arc length. The axial strain from the ground is not transferred to the cable and cannot be measured by DAS.

deformation U_l , and the bending energy U_f are (e.g., Weaver et al., 1990)

$$U_l = \frac{1}{2} \frac{EA}{L} q_l^2, \quad \& \quad U_f = \frac{1}{2} EIC_2 q_f^2, \quad C_2 = \int_0^L (\phi''(x))^2 dx = \frac{2\pi^4}{L^3}. \quad (5)$$

The total deformation energy is then

$$U = U_l + U_f = \frac{1}{2} \frac{EA}{L} q_l^2 + \frac{1}{2} EIC_2 q_f^2. \quad (6)$$

The total potential energy of the cable under the load of gravity is $\Pi = U - W$, where

W is the work done by gravity. Assuming that the cable is under no initial axial tension $U_l = 0$ and $\Pi = U_f - W$. Gravity acts as a uniform distributed load ρAg along the segment, doing work

$$W = \int_0^L \rho Ag w(x) dx = \frac{1}{2} \rho Ag L q_f = \frac{1}{2} F_g q_f \quad (7)$$

as the segment sags, where $F_g = \rho Ag L$ is the total gravitational force on the segment. Equilibrium occurs at the minimum of the potential energy ($\frac{d\Pi}{dq_f} = 0$):

$$\frac{d}{dq_f} \left(\frac{1}{2} EIC_2 q_f^2 - \frac{1}{2} F_g q_f \right) = EIC_2 q_f - \frac{1}{2} F_g = 0. \quad (8)$$

This yields the equilibrium condition where the internal elastic restoring force matches the external gravitational force, resulting in an expression of the gravity-induced initial sag q_{f0} :

$$EIC_2 q_{f0} = \frac{1}{2} F_g \Rightarrow q_{f0} = \frac{\rho Ag L}{2EIC_2} = \frac{\rho Ag L^4}{4\pi^4 EI}. \quad (9)$$

We consider a small ground movement δL , which causes small changes $q_f = q_{f0} + \delta q_f$ and $q_l = \delta q_l$. Linearizing about the initial state ($q_l = 0, q_f = q_{f0}$):

$$L = L_0 + q_l - \frac{1}{2} C_1 q_f^2 \Rightarrow \delta L = \delta q_l - C_1 q_{f0} \delta q_f \quad (10)$$

We minimize the incremental energy δU required to accommodate the ground movement δL :

$$\delta U = \frac{1}{2} \frac{EA}{L} (\delta q_l)^2 + \frac{1}{2} EIC_2 (\delta q_f)^2. \quad (11)$$

Substitute the expression for δq_l from the linearized constraint:

$$\delta U = \frac{1}{2} \frac{EA}{L} (\delta L + C_1 q_{f0} \delta q_f)^2 + \frac{1}{2} EIC_2 (\delta q_f)^2. \quad (12)$$

We minimize δU with respect to δq_f and solve for δq_f :

$$\delta q_f = - \frac{\frac{EA}{L} C_1 q_{f0} \delta L}{EIC_2 + \frac{EA}{L} (C_1 q_{f0})^2} \quad (13)$$

Then, plug δq_f back into the constraint:

$$\delta q_l = \delta L + C_1 q_{f0} \delta q_f = \delta L \frac{EIC_2}{EIC_2 + \frac{EA}{L} (C_1 q_{f0})^2}. \quad (14)$$

This yields the axial strain transfer ratio,

$$\frac{\delta q_l}{\delta L} = \frac{1}{1 + \Theta}, \quad (15)$$

with

$$\Theta = \frac{EA}{L} \frac{(C_1 q_{f0})^2}{EIC_2} = \frac{\rho^2 g^2 A^3 L^8}{128 \pi^8 E^2 I^3}. \quad (16)$$

Equation 16 expresses Θ in terms of the cable’s mechanical and geometric parameters. Inserting the initial sag $w_0(L/2) = q_{f0}$ (Equation 9), and using $A = \pi r^2$ and $I = \pi r^4/4$, the parameter reduces to the compact, physically interpretable form

$$\Theta = \frac{1}{2} \left(\frac{w_0(L/2)}{r} \right)^2. \quad (17)$$

The static strain transfer efficiency thus depends entirely on the ratio of the segment’s initial sag to its radius: small sag relative to the cable’s thickness yields near-complete axial strain transfer, while large sag compared to the radius causes the segment to accommodate endpoint displacement through bending rather than through axial elongation.

3 Numerical Methods

To numerically evaluate strain transfer, we model the suspended cable segment as a 1D Timoshenko beam. This theoretical formulation assumes linear elasticity and incorporates finite rotations and shear, allowing for the accurate simulation of large geometric deformations and coupling between transverse bending and axial elongation.

The beam model is implemented using the open-source finite element framework FEniCS. The numerical solution is obtained in two primary steps: initially, a nonlinear Newton solver is used to determine the static equilibrium shape of the beam under gravity, for two deployment scenarios where the endpoints are either clamped with initial tension or permitted to relax horizontally (as assumed in the analytical solution). Following this static step, we linearize around the gravity-sagged equilibrium position and analyze the dynamic behavior by solving for the natural vibration modes and performing a steady-state frequency response analysis. Dissipation is modeled with a constant structural (hysteretic) damping, implemented as a complex stiffness $K(1 + i\eta)$ with loss factor $\eta = 0.1$.

The mechanical and geometric parameters which govern the behavior of the Timoshenko beam, together with the ranges over which they were varied, are listed in Table 1. Each parameter was sampled with geometric (logarithmically even) spacing between its minimum and maximum values. We chose this spacing because the strain transfer collapses onto the dimensionless parameter Θ , which depends on the input parameters through a product of power laws. Equal multiplicative steps therefore give even coverage in Θ . The suspended-segment length L , to which Θ is most sensitive ($\Theta \propto L^8$), was sampled at five points, and the remaining parameters at three. We modeled all combinations of the sampled values to evaluate their collective influence on the coupling behavior.

Table 1: Parameters governing the behavior of a cable segment modeled as a Timoshenko beam: mechanical properties (Young’s modulus E , shear modulus G , density ρ), geometric parameters (radius r , segment length L), and gravitational acceleration g . Each parameter was sampled with geometric spacing between its minimum and maximum values, at the number of points given in the last column. The shear modulus G enters the numerical model because it accounts for shear, but cables are slender beams for which shear is negligible, so varying G has very little effect on the results.

Parameter	Symbol	Unit	Min.	Max.	Points
Cable radius	r	mm	0.1	2.5	3
Segment length	L	cm	5	20	5
Young’s modulus	E	GPa	0.1	10	3
Shear modulus	G	GPa	0.05	5	3
Density	ρ	kg m ⁻³	500	1500	3
Gravitational acceleration	g	m s ⁻²	1	10	3

4 Results

4.1 Initial Cable Sag Under Gravity

The initial sag of a suspended cable segment under gravity, given analytically by Equation 9, is governed by the balance between the downward pull of gravity and the cable’s inherent bending stiffness.

The numerical simulations confirm this: results agree closely with the linear analytical model at small to moderate sags Figure 2a. At the smallest sags, the numerical results sag slightly more than the analytical prediction. These cases correspond to segments that are short relative to their radius, for which shear deformation is no longer negligible. The Timoshenko beam used numerically accounts for this shearing, whereas the analytical Euler-Bernoulli model does not, so the analytical model slightly underpredicts the sag. On the other hand, at very large dimensionless ratios, representing highly compliant or long free-hanging segments, the numerical results diverge from the linear prediction because the small-sag approximation made in the analytical model breaks down. Furthermore, the numerical results demonstrate that an initial axial tension resists transverse deflection, restricting the initial sag to values lower than predicted by the tension-free analytical model.

4.2 Static Strain Transfer Efficiency

The numerical simulations validate the analytical prediction that strain transfer efficiency decays rapidly for segments characterized by large Θ values (Figure 2b). When the initial sag is much smaller than the cable’s radius ($w_0(L/2) \ll r$), the segment behaves like a straight rod, and the boundary displacement is transferred almost entirely as axial strain. As Θ increases, the initial sag increases and the segment accommodates a growing fraction of the endpoint displacement by changing its bending rather than its length, and the strain transfer efficiency decreases gradually. As a practical threshold, keeping the loss in transfer efficiency below 3% requires $\Theta < 1/32$ corresponding to a sag smaller than about one quarter of the cable radius ($w_0(L/2)/r < 1/4$). Notably, if the specific deployment configuration includes an initial axial pre-tension, the numerical simulations show that

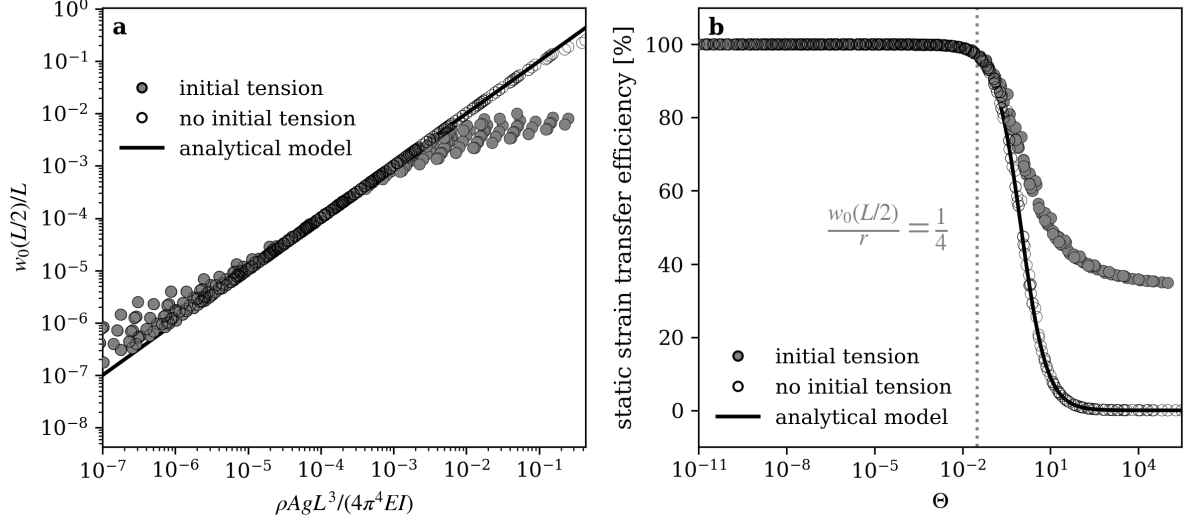


Figure 2: **Static behavior of an unburied cable segment under gravity.** The initial cable sag relative to the suspended segment length scales linearly with the ratio of gravitational to bending resisting forces, while the resulting static strain transfer efficiency is dictated by the dimensionless parameter Θ . **(a)** Initial dimensionless midpoint sag ($w_0(L/2)/L$) as a function of the force ratio. **(b)** Static strain transfer efficiency as a function of Θ , which depends on the ratio of the cable’s initial sag to its radius. Solid lines indicate analytical predictions from equations derived in Section 2.2, while markers represent numerical results. At small Θ , the cable behaves like a straight rod, yielding near-perfect axial strain transfer. At large Θ , the cable behaves like a hanging string, transforming axial boundary displacements into changes in sag instead of transferring axial strain to the fiber. This efficiency loss at large Θ values is less severe for deployments with initial axial tension.

the strain transfer drop-off is less pronounced. Instead of decaying towards zero at large Θ regimes, the initial tension restricts the amount of initial sag, which results in a higher axial strain transfer efficiency.

4.3 Dynamic Strain Transfer Efficiency

Natural Resonance Frequency

The dynamic response of a suspended cable segment depends on its natural frequencies, most importantly the first flexural bending mode. This fundamental resonance frequency acts as a threshold dividing the cable’s behavior into quasi-static, resonant, and inertia-dominated regimes. For a straight, horizontal Euler-Bernoulli beam with doubly clamped boundary conditions, the fundamental angular frequency is (e.g., Weaver et al., 1990)

$$\omega_1 = \frac{22.4}{L^2} \sqrt{\frac{EI}{\rho A}} \quad (18)$$

If the seismic frequency band of interest lies sufficiently far below this resonance, the cable segment behaves in a predictable, frequency-independent manner.

While the fundamental frequency depends primarily on these inherent properties, gravity-induced sagging causes a geometric stiffening effect that increases the resonance frequency.

A cable pre-bent by its own weight naturally exhibits a higher effective flexural stiffness than a perfectly straight counterpart. However, this gravity-induced frequency shift is generally of secondary importance; the absolute magnitude of the shift remains small compared to the variations resulting from the primary geometric and mechanical properties of the cable.

Dynamic Cable Response

To examine the dynamic response across the range of Θ , we consider a single cable of fixed material and cross-section and vary the suspended segment length L (Figure 3). Increasing the segment length increases Θ and, at the same time, lowers the fundamental resonance frequency through Equation 18, as seen in the leftward shift of the resonance peaks with increasing Θ in Figure 3.

At seismic frequencies well below the fundamental resonance, the cable responds quasi-statically, yielding a dynamic strain transfer efficiency that matches the static limit (Figure 3c). Because the driving ground motion is slow compared to the cable’s natural mechanical response time, internal inertial forces remain negligible, and the segment is able to transition between equilibrium states in phase with the ground.

How strongly the segment deviates from its quasi-static limit as frequency increases depends itself on Θ . Below $\Theta \approx 1/32$, the strain transfer remains close to 100% across the entire frequency band, and the amplitude and phase responses stay essentially flat (Figure 3c and d). For larger Θ , the response becomes increasingly frequency-dependent, with strong deviations from the static limit that we describe in the following.

As the driving frequency increases toward the fundamental resonance, the ground strain strongly couples into the transverse bending mode, resulting in a peak of the midpoint displacement amplitude at the resonance frequency. Before reaching the resonance, the midpoint deflection is still close to in-phase with the ground movement. Consequently, ground displacement is absorbed into bending more effectively. This destructive coupling causes the strain transfer efficiency to drop below its static limit and reach a minimum just before the resonance frequency.

Traversing the resonance frequency the phase of the midpoint displacement changes to an almost 180-degree phase shift, inverting how the bending mode and the axial mode interact. As the phase shifts past -90 degrees toward fully out-of-phase (-180 degrees), the cable starts moving in opposition to the ground driving motion: when the endpoints are pulled outward, the inertia of the cable causes it to bend further, rather than to straighten. The two effects add constructively: the fiber is stretched both by the ground and by its own whipping motion. This amplifies the signal, causing the strain transfer efficiency to spike to well over 100% immediately above the resonance frequency. The severity of this amplification increases with Θ (Figure 3c).

At driving frequencies significantly exceeding the first flexural mode, structural inertia ultimately suppresses the bending mode. The transverse flexure becomes “too slow” to follow the high-frequency oscillations, causing the midpoint displacement to decay toward zero. While the much stiffer axial mode is still able to respond to the driving motion, the coupling between the modes becomes negligible. The cable segment is effectively forced into a purely axial deformation regime. Consequently, independent of Θ , the dynamic strain transfer efficiency approaches 100% past the resonance frequency.

At even higher frequencies, the higher flexural modes of the segment produce additional, weaker peaks and phase shifts (Figure 3).

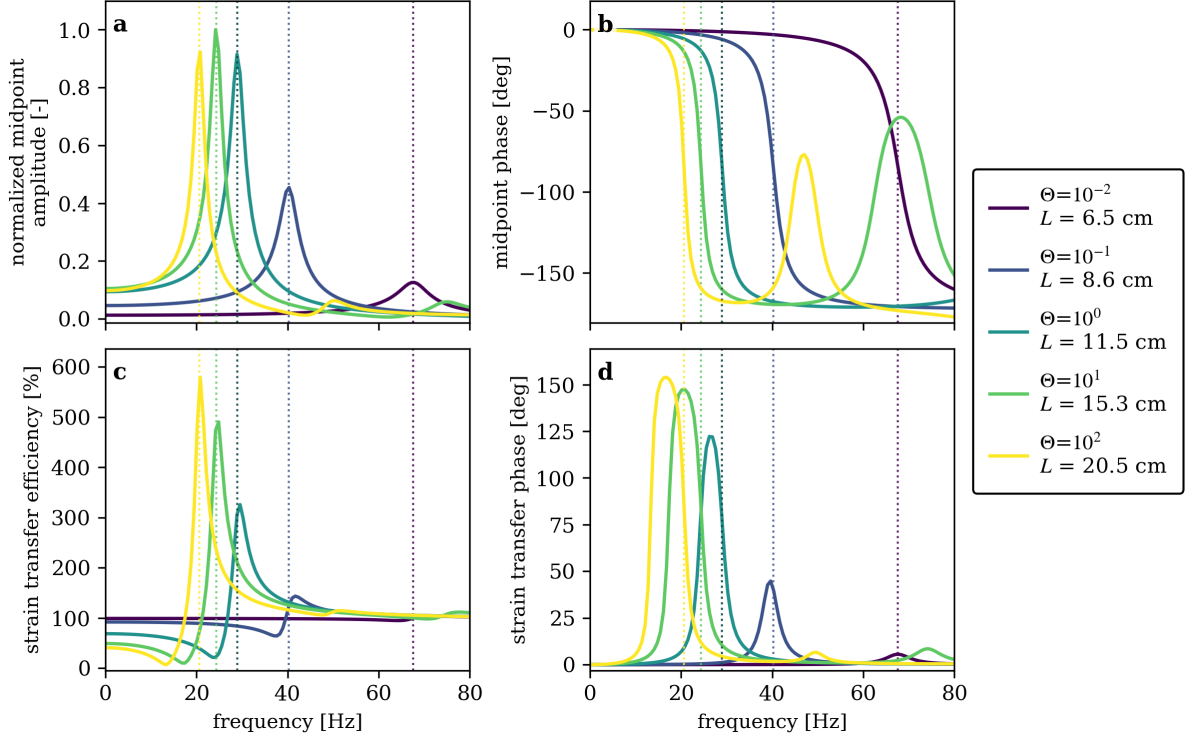


Figure 3: Dynamic behavior of the axial and bending mode coupling and the resulting strain transfer efficiency. Frequency response of a single suspended cable segment as the segment length L is varied to span a range of Θ , at fixed gravity ($g = 9.81 \text{ m s}^{-2}$) and fixed cable properties ($E = 0.1 \text{ GPa}$, $\rho = 1000 \text{ kg m}^{-3}$, $r = 0.5 \text{ mm}$). The corresponding segment lengths are given in the legend. The segment is clamped with an initial axial tension. **(a, b)** Amplitude and phase of the cable's midpoint deflection relative to its sagged state (not absolute deflection), measuring the intensity of the flexural mode during axial deformation. The deflection shows a typical resonance peak and a frequency-dependent phase shift. **(c, d)** Amplitude and phase of the dynamic strain transfer efficiency as a function of frequency. At low frequencies, the strain transfer matches the static limit. As the frequency approaches resonance, efficiency is initially attenuated due to destructive mode coupling, followed by an amplification peak driven by constructive coupling as the phase rotates across the resonance. Well above the resonance frequency, cable inertia suppresses the bending mode entirely, resulting in near 100% strain transfer efficiency. Higher order deformation modes produce additional peaks at higher frequencies. Increasing Θ here corresponds to a longer segment, which lowers the resonance frequency (Equation 18), shifting the peaks to the left.

5 Discussion

Sag-to-radius ratio governs strain transfer

The strain transfer efficiency of a suspended cable segment is determined by a single dimensionless parameter, the ratio of its gravity-induced sag to its radius, captured by Θ . A segment with small sag relative to its radius is mechanically close to a straight rod: lacking a sagging geometry to exploit, it can only accommodate an endpoint displacement through changing its length, and the ground strain is transferred almost entirely as axial strain to the fiber. A segment with large sag instead accommodates the same displacement

by changing the amount of bending, absorbing the strain instead of transferring it to the fiber.

The transition between these regimes is gradual: the strain transfer decreases as Θ exceeds about $1/32$, corresponding to a sag of a quarter of the cable radius. Introducing an initial axial tension raises the efficiency at a given Θ by restricting the sag, but the value of Θ at which the efficiency begins to drop off is unchanged; in either case, the objective for a reliable deployment is to keep Θ as small as possible.

This mechanism offers a possible explanation for empirical observations that have so far lacked a mechanical interpretation. Because a draped cable sags between contact points and absorbs part of the ground strain as bending, our model predicts that unburied cables record lower amplitudes than buried ones, and that this loss is reduced for thicker and stiffer cables, which sag less. These predictions are consistent with the reduced amplitudes reported for unburied deployments (An et al., 2023; Harmon et al., 2022; Zandanel et al., 2026), and with the observation that thicker and stiffer cables couple more reliably (Probst et al., 2026), although other factors may also contribute. A direct quantitative comparison is not possible, however, since evaluating Θ requires the characteristic segment length L , which none of these studies measured or reported.

All of this assumes the cable responds quasi-statically, which requires the seismic frequencies of interest to lie well below the segment’s fundamental resonance. Near resonance, the strain transfer efficiency deviates strongly from the static limit, with large amplitude variations and phase shifts that make the recorded strain unreliable and difficult to correct. This regime must therefore be avoided. The resonance frequency and Θ share a common dependence on the cable’s stiffness, density, and geometry, so for a given gravity, a cable that is thicker, stiffer, or spans shorter segments has both a lower Θ and a higher resonance frequency, pushing the resonance further above the seismic band. A deployment optimized for low Θ therefore also avoids the problematic dynamic regime.

The strain transfer derived here applies to a single segment, whereas a DAS measurement averages over many segments of varying length within each gauge length. In the rod-like regime this averaging has little effect, since every segment transfers strain efficiently. However, as segments move into the bending-dominated regime, the transferred strain depends strongly on segment length, and the gauge-length response becomes a weighted average over an unknown distribution of segment lengths. Recovering the true ground strain by calibration then becomes impractical, which is another reason to keep deployments within the rod-like regime.

Bending stress relief sets an upper bound on strain transfer

Bending stress relief, the mechanism we have identified and quantified here, may not be the only one limiting coupling in unburied deployments. Other mechanisms can act alongside it, including the frictional slip at the contact points introduced earlier. Where such mechanisms are active, they would further reduce the transferred strain below the values predicted here, so our model gives an upper bound on the achievable strain transfer. Our analysis also addresses only the first of the two transfer steps. The efficiency derived here concerns only the ground-to-cable transfer, and we have assumed that the strain is then transmitted perfectly through the cable to the fiber. In practice this cable-to-fiber transfer can be imperfect, attenuating the strain through shear-lag mechanics, though it approaches unity for suitable cable constructions (section 1). The strain reaching the fiber is therefore at most that predicted by Θ , and may be lower depending on the cable.

Lower lunar gravity reduces bending stress relief but raises slip risk

The reduced gravity on the Moon has two opposing effects on coupling. Because Θ scales with the square of gravity, the roughly six times lower lunar gravity reduces the gravity-induced sag, favoring efficient strain transfer. At the same time, the lower gravity reduces the normal force at the contact points and thus the static friction available to resist sliding, making frictional slip more likely.

Whether slip is actually triggered depends on the amplitude of the ground motion, since slip occurs only once the induced axial force exceeds the available friction. Seismic events on the Moon are generally much weaker than on Earth, so the axial forces they induce at the contact points are correspondingly small and may remain below the friction threshold even at the reduced lunar normal force. The increased slip risk from lower gravity may therefore be mitigated, at least for the weak ground motions expected from passive lunar sources. Stronger sources, such as active surveys, could still exceed it.

Limitations of the model

The model assumes that each cable segment is initially straight, with no inherent curvature. In practice, cables retain some bending from being stored on a spool, which adds to the gravity-induced sag. Because the strain transfer is controlled by the total sag relative to the radius, any inherent bending increases the effective Θ and lowers the strain transfer below the values predicted here. This effect is likely significant for typical cables and is the most important deviation of real deployments from the idealized model.

The model further assumes that the contact points move identically to the surrounding subsurface. This is a prerequisite for the model to apply, and it holds for a cable resting on compact soil or rock. On vegetated ground it does not: compliant material such as grass deforms under the cable, so the contact points no longer follow the bulk ground motion, and the model no longer describes the coupling. Reliable strain transfer in an unburied deployment thus requires direct contact with firm ground.

The analysis considers only inline ground strain, aligned with the cable axis, because this is what DAS predominantly senses: in a well-coupled deployment, DAS is largely insensitive to strain perpendicular to the cable. In an unburied deployment, however, broadside motion may couple into the axial response through the segment's dynamics. If both endpoints of a suspended segment are displaced transversely, the inertia of the cable resists the motion and stretches the segment slightly, producing a small axial strain. This effect is likely minor, but it is not captured by the present model and would require accounting for non-inline incidence.

Finally, the frequency response analysis assumes steady-state harmonic excitation, whereas seismic signals are transient. This assumption affects primarily the behavior near resonance. Since reliable measurements require operating well below resonance in any case, the transient response does not affect the practical conclusions drawn here.

Design and deployment guidelines for minimizing Θ

The results translate into concrete guidelines for unburied deployments, all following from the goal of keeping Θ , and thus the gravity-induced sag relative to the radius, as small as possible. The cable itself should have a large radius and high stiffness while remaining lightweight, since Θ decreases with radius and stiffness and increases with density.

Stiffness should not be achieved through rigid metallic construction, however, as such cables have been observed to transmit sound along their length and may act as acoustic waveguides that introduce artifacts (Carr et al., 2025).

The cable should also not have any inherent curvature, as an initial bend adds to the sag

and reduces strain transfer. Cables that maintain a bend from being stored on a spool should therefore be avoided.

On the deployment side, the suspended segment length matters most, on which Θ depends very strongly (as L^8). Maximizing ground contact, by deploying on flat surfaces and minimizing surface roughness so that free-hanging segments stay short, is therefore essential. For a lunar deployment, the weight of the cable should be minimized, since launch mass is very costly. This creates a tradeoff, as a larger radius and stiffer materials tend to increase weight. Suitable cables should therefore be stiff and thick without being heavy.

6 Conclusion

We modeled unburied, surface-draped DAS cables as a sequence of beam segments suspended between discrete ground contact points, and derived analytical and numerical solutions for the axial strain transferred from the ground to the cable. The strain transfer efficiency is governed by a single dimensionless parameter, $\Theta = \frac{1}{2} (w_0(L/2)/r)^2$, determined fully by the ratio of the segment's gravity-induced sag to its radius. When the sag is small relative to the radius ($w_0(L/2)/r < 1/4$, i.e., $\Theta < 1/32$), the segment behaves like a straight rod and transfers ground strain almost completely to the fiber. As the sag grows, the segment increasingly takes up endpoint displacement by changing its curvature rather than its arc length, so that less of the ground strain reaches the fiber as measurable axial strain.

This is the first quantitative framework for ground-to-cable coupling in unburied DAS, and it offers a predictive criterion for when a given deployment will couple well. Because any additional mechanism, such as frictional slip, can only reduce transfer further, Θ sets an upper bound on the achievable strain transfer. Coupling improves for cables with a larger radius, higher stiffness, and lower density, and for shorter segments between contact points. On the Moon, the roughly six times lower gravity reduces the sag and improves the strain transfer, but it also lowers the normal force at the contact points, and increases the risk of frictional slip. Validating the model experimentally and understanding how inherent cable bending and frictional slip affect the transfer are the next steps toward a complete picture of unburied DAS coupling and a basis for optimizing deployments on both Earth and the Moon.

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